**Q1.**

Let the distance between the foci of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be 40. If its eccentricity is the minimum value of the function $g(t) = \sec^2 t + \csc^2 t, t \in \left(0, \frac{\pi}{2}\right)$, then the length of its latus rectum is equal to -

(1) 150 (2) 160 (3) 180 (4) 200

**Q2.**

A line $L$ with negative slope passes through $P(1, 3)$ and meets the positive coordinate axes at $A$ and $B$. Let $S$ be the minimum possible area of $\Delta OAB$ (where $O$ is the origin). If $\theta$ is the angle $L$ makes with the positive x-axis when the area is minimum, the value of $\frac{S}{|\sin 2\theta|}$ is:

(1) 8 (2) 10 (3) 12 (4) 15

**Q3.**

Let $I = \int_{-1}^{1} \left( \frac{x^7 - 3x^5}{1 + \cos^2(\pi x)} + 2 \right) dx$.

If $M = \begin{bmatrix} I & 1 & 2 \\ 0 & 1 & 1 \\ \alpha & -1 & 1 \end{bmatrix}$ and $B = \text{adj}(2M)$ be such that $|B| = 1024$, then the sum of all possible values of $\alpha$ is equal to

(1) 8 (2) 12 (3) 16 (4) 20

**Q4.**

Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$. A vector $\vec{c}$ satisfies the equations $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 9$. If $d$ is the perpendicular distance from the point with position vector $\vec{c}$ to the line passing through the origin and parallel to the vector $\vec{a} \times \vec{b}$, then $d^2$ is equal to:

(1) 3 (2) 9 (3) 27 (4) 81

**Q5.**

Let $P(x) = x^3 + px^2 + qx + r$ be a polynomial whose roots are three distinct positive integers forming an Arithmetic Progression (A.P.). If $P(0) = -105$ and $p = -15$, then the value of $P(2)$ is equal to:

(1) $-15$ (2) 15 (3) $-9$ (4) 9

**Q6.**

Let $A$ be an invertible square matrix of order 3 such that $A(\operatorname{adj}(2A)) = 32I$, where $I$ is the identity matrix of order 3. If $B = \operatorname{adj}(A) - 4A^{-1}$, then the value of the determinant of $B$ is:

(1) 64 (2) 56 (3) 16 (4) 8

**Q7.**

A parabola has focus $S(2,3)$ and directrix $x+y+1=0$. If its vertex is $V(\alpha,\beta)$ and the length of its latus rectum is $L$, then $2(\alpha+\beta)+L^2$ is equal to

(1) 72 (2) 74 (3) 76 (4) 78

**Q8.**

The sum of the infinite series
$$1\cdot\left(\frac{1}{3}+\frac{1}{4}\right)+2\cdot\left(\frac{1}{3^2}+\frac{1}{3\times4}+\frac{1}{4^2}\right)+3\cdot\left(\frac{1}{3^3}+\frac{1}{3^2\times4}+\frac{1}{3\times4^2}+\frac{1}{4^3}\right)+\cdots$$

(1) $\frac{4}{3}$ (2) $\frac{5}{3}$ (3) $\frac{7}{4}$ (4) $\frac{5}{2}$

**Q9.**

The letters of the word "BHAWAN" are written in all possible ways with or without meaning and these words are arranged as in a dictionary. The number of words appearing strictly between the words "BHAWAN" and "NAWABH" is:

(1) 108 (2) 109 (3) 110 (4) 111

**Q10.**

The largest value of $k$, for which $45^k$ divides $1\cdot3\cdot5\cdots149$, is

(1) 19 (2) 17 (3) 18 (4) 20

**Q11.**

Let the lines $L_1 : \dfrac{x-1}{1}=\dfrac{y-2}{2}=\dfrac{z+1}{3}$ and $L_2 : \dfrac{x-2}{2}=\dfrac{y-4}{3}=\dfrac{z-\lambda}{4}$ intersect each other. If $d$ is the shortest distance between the line $L_2$ and the z-axis, then $13d^2$ is equal to -

(1) 2 (2) 4 (3) 6 (4) 8

**Q12.**

Let the mean and variance of 6 observations be 4 and 5 respectively. If two new observations $\alpha$ and $\beta$ are added to this data, the mean of the 8 observations becomes 5 and their variance becomes 10. The quadratic equation whose roots are $\alpha$ and $\beta$ is -

(1) $x^2 - 16x + 51 = 0$
(2) $x^2 - 16x + 3 = 0$
(3) $x^2 - 10x + 51 = 0$
(4) $x^2 - 8x + 56 = 0$

**Q13.**

If the domain of the function $f(x) = \cos^{-1}\left(\frac{x^2 - 2x}{x^2 - 4}\right)$ is $[\alpha, \beta) \cup (\gamma, \infty)$, then $\alpha + \beta + \gamma$ is equal to

(1) 1
(2) 2
(3) 3
(4) 5

**Q14.**

Consider the following three statements for the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = |x^2 - 2x|e^{-x}$:
(I) $f$ is non-differentiable at exactly two points.
(II) $f$ is strictly decreasing in the interval $(1, 2)$.
(III) $f$ has a local minimum at $x = 2 + \sqrt{2}$.
Then,

(1) All (I), (II) and (III) are TRUE.
(2) Only (I) and (II) are TRUE.
(3) Only (II) and (III) are TRUE.
(4) Only (I) and (III) are TRUE.

**Q15.**

Let $f$ be a function such that $3f(x) - 2f(2 - x) = x^2 + \lambda x$ for all real $x$, where $\lambda = \sum_{n=1}^{\infty} \frac{6}{3^n}$. Then the value of $\sum_{i=1}^{5} f(i)$ is equal to -

(1) 45
(2) 50
(3) 60
(4) 75

**Q16.**

Let $y = y(x)$ be a differentiable function in the interval $(0, \infty)$ satisfying the integral equation $x^2 y(x) - \int_1^x t y(t)\, dt = x^3 + 1$. If $y(1) = 2$, then the value of $4y(2)$ is equal to

(1) 9
(2) 13
(3) 17
(4) 21

**Q17.**

Let $A_k$ denote the area of the region bounded by the parabola $y = x^2$ and the curve $y = k|x|$. If $A_m - A_n = 21$ for some positive integers $m > n$, then the value of $(m^2 + n^2)$ is equal to:

(1) 15 (2) 17 (3) 20 (4) 25

**Q18.**

The smallest positive integral value of $p$, for which exactly one root of the equation $x^2 - px + 15 = 0$ lies in the interval $(2, 4)$, is equal to

(1) 7 (2) 8 (3) 9 (4) 10

**Q19.**

Let $[t]$ denote the greatest integer less than or equal to $t$. If the function

$$f(x) = \begin{cases} p^2\left(1 + \left[\dfrac{3\sin x}{x}\right] - \left[\dfrac{2\tan x}{x}\right]\right) & , x < 0 \\ \dfrac{1}{x^2}\int_0^x \dfrac{e^{t^2} - 1}{t}\,dt & , x > 0 \\ q & , x = 0 \end{cases}$$

is continuous at $x = 0$, then the value of $(4p^2 + 8q)$ is equal to:

(1) 5 (2) 6 (3) 4 (4) 8

**Q20.**

Let $\vec{a} = \hat{i} + \hat{j} - \hat{k}$, $\vec{b} = -\hat{i} + \hat{j}$ and $\vec{c} = \hat{i} + 5\hat{j} + 4\hat{k}$. If $\vec{x}$ and $\vec{y}$ are two vectors satisfying the equations $\vec{x} + \vec{y} \times \vec{a} = \vec{b}$ and $\vec{y} + \vec{x} \times \vec{a} = \vec{c}$, then the value of $|\vec{x} \times \vec{y}|^2$ is equal to -

(1) 15 (2) 25 (3) 35 (4) 45

**Q21.**

Let $(h, k)$ lie on the circle $C : x^2 + y^2 = 16$ and the point $(2h - 1, k^2 + 2)$ lie on a parabola with latus rectum $L$. Then the value of $5L$ is equal to

**Q22.**

The number of elements in the set
$\{x \in [0, 180^\circ] : 4\sin(x)\sin(60^\circ - x)\sin(60^\circ + x) = \sin(2x)\}$ is ____.

**Q23.**

Let $S$ be a set of 4 elements and $P(S)$ denote its power set. Let an ordered pair $(A, B)$ be chosen randomly from $P(S) \times P(S)$ given that $A \cup B = S$. If the probability that $|A \cap B| = 1$ is $\frac{p}{q}$, where $p$ and $q$ are coprime, then $p + q$ is equal to ______

(Note: $|E|$ represents the number of elements in set $E$.)

**Q24.**

Let $z$ be a complex number such that $z + |z| = 2 + 8i$, where $i = \sqrt{-1}$. Then the value of $|z|^2$ is equal to

**Q25.**

If $I = \int_0^{2026} \frac{\sqrt{x}}{\sqrt{x} + \sqrt{2026 - x}}\, dx$ and $J = \int_{-1}^{1} \frac{x^2}{1 + 2026^x}\, dx$, then the value of $I + 3J$ is equal to -

**Q26.**

A screw gauge (pitch 0.5 mm, 50 circular divisions) has a positive zero error of 5 circular scale divisions. If a wire's measurement shows a 2 mm main scale reading and the 30th circular division coinciding, its true diameter is:

(1) 2.35 mm (2) 2.25 mm (3) 2.30 mm (4) 1.95 mm

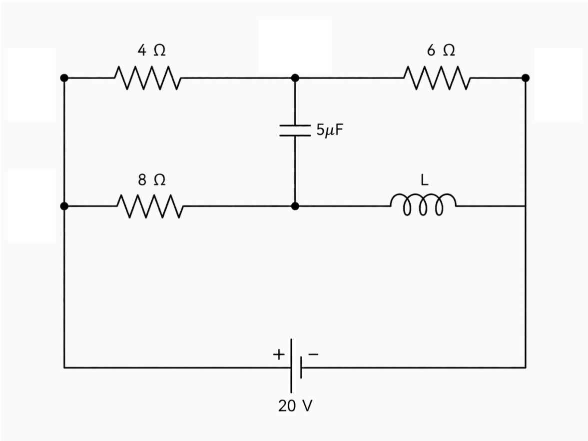
**Q27.**

A glass biconvex lens ($\mu_g = 1.5$) has a focal length of 20 cm in air. The entire optical setup is completely immersed in a transparent liquid ($\mu_l = 1.2$). Four students, $A$, $B$, $C$, and $D$, attempt to form a real image of an object on a screen using this immersed lens setup. They record the distance between the object and the screen ($D$) as follows:
Student A: 130 cm
Student B: 150 cm
Student C: 170 cm
Student D: 190 cm
Which of the following options correctly identifies all the students who recorded physically impossible object-to-screen distances?

(1) Only Student A
(2) Student A and Student B
(3) Student C and Student D
(4) Student A, Student B, and Student C

**Q28.**

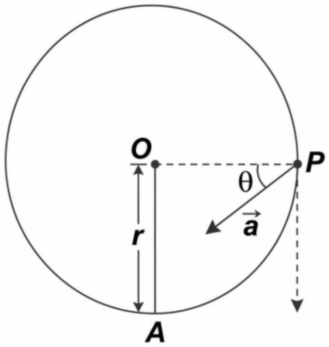
Find the steady-state energy stored in the capacitor for the given circuit (assuming an ideal inductor and battery).




(1) 160 $\mu$J (2) 360 $\mu$J (3) 720 $\mu$J (4) 0 $\mu$J

**Q29.**

In the given figure, a particle attached to a string of length $r$ is projected from the lowest point A. When the string is horizontal, the net acceleration vector makes an angle $\theta$ with the horizontal. If $\tan\theta = 1/3$, the velocity at A is:




(1) $\sqrt{3gr}$ (2) $\sqrt{4gr}$ (3) $\sqrt{5gr}$ (4) $\sqrt{7gr}$

**Q30.**

Two identical circular coils, $M$ and $N$, each of radius $R$, share a common center $O$ but lie in mutually perpendicular planes. They carry currents $I$ and $\sqrt{3}I$ respectively. A particle of charge $+q$ passes through $O$ with velocity $v$ directed strictly along the central axis of coil $M$. The magnitude of the instantaneous magnetic force on the particle is

(1) 0 (2) $\dfrac{\mu_0 qvI}{2R}$ (3) $\dfrac{\sqrt{3}\mu_0 qvI}{2R}$ (4) $\dfrac{\mu_0 qvI}{R}$

**Q31.**

A composite rigid body is formed by a slender uniform thin rod (mass $M$, length $L$) and an attached small point mass (mass $M$) at one of its ends. It is pivoted at the other rod-only end. If held initially in an upright vertical position (with the attached mass on top) and then released, what is the system's angular velocity as it sweeps past the downward vertical position? (Gravitational acceleration is $g$).

(1) $2\sqrt{g/L}$ (2) $\frac{3}{\sqrt{2}}\sqrt{g/L}$ (3) $\sqrt{2g/L}$ (4) $\sqrt{3g/L}$

**Q32.**

A metallic surface having a work function $\Phi$ is illuminated by photons of wavelength $\lambda$. The de Broglie wavelength of the fastest emitted photoelectrons is found to be $\lambda_d$. If the incident light is replaced by another light of wavelength $\lambda/2$, the new de Broglie wavelength $\lambda_d'$ of the fastest photoelectrons satisfies:

(1) $\lambda_d' = \frac{\lambda_d}{\sqrt{2}}$ (2) $\lambda_d' < \frac{\lambda_d}{\sqrt{2}}$
(3) $\lambda_d' > \frac{\lambda_d}{\sqrt{2}}$ (4) $\lambda_d' = \frac{\lambda_d}{2}$

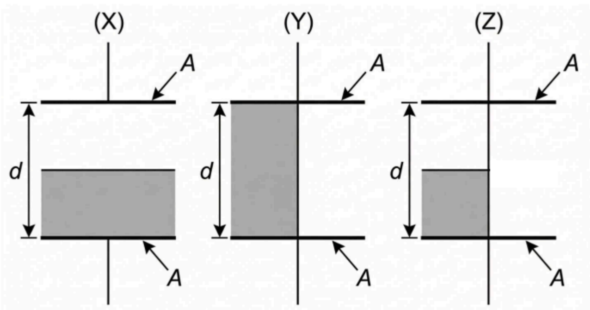
**Q33.**

The first overtone of an open air pipe is in unison with the second overtone of a closed air pipe. If the closed pipe is refilled with a gas where the speed of sound is doubled, its new fundamental frequency becomes $k$ times the open pipe's fundamental frequency. The value of $k$ is

(1) 0.4 (2) 1.25 (3) 0.8 (4) 1.0

**Q34.**

Three isolated identical capacitors, each carrying charge $Q$, are partially filled with a dielectric ($K = 2$). $X$ is filled for half its thickness. $Y$ is filled for half its area. $Z$ is filled for half its thickness and half its area. Which is true for their final stored energies?




(1) $U_Y > U_X > U_Z$ (2) $U_Z > U_X > U_Y$
(3) $U_X > U_Y > U_Z$ (4) $U_Z > U_Y > U_X$

**Q35.**

A regular square network is formed by four outer wires each of resistance $r\ \Omega$, and the corners are joined to the centre by wires each of the same resistance, $r\ \Omega$. A cell of internal resistance $r/3\ \Omega$ is connected across two diagonally opposite corners. The ratio of total thermal power dissipated in the wire network to that dissipated inside the cell is:

(1) 1 : 2 (2) 2 : 3 (3) 2 : 1 (4) 3 : 2

**Q36.**

A solid body floats in water with 40% of its volume above the surface. When placed in an oil, it floats with 60% of its volume above the surface. The specific gravity of the oil is:

(1) 0.67 (2) 1.5 (3) 0.8 (4) 1.2

**Q37.**

A stationary nucleus $X$ ($A = 240$) undergoes fission into fragments $Y$ ($A = 100$) and $Z$ ($A = 140$). The binding energy per nucleon of $X$ is 7.5 MeV and of both products is 8.5 MeV. Assuming all released energy becomes kinetic energy, the ratio of kinetic energy of $Y$ to $Z$ is:

(1) 5 : 7 (2) 7 : 5 (3) 1 : 1 (4) 25 : 49

**Q38.**

A galvanometer has a coil resistance of 50 Ω and shows full-scale deflection for a current of 2 mA. It is converted into an ammeter to measure currents up to 10 mA by connecting a suitable shunt. The effective resistance of the resulting ammeter is:

(1) 12.5 Ω (2) 10 Ω (3) 8 Ω (4) 40 Ω

**Q39.**

A small sphere of mass $m$ is suspended by a light string from a fixed ceiling. A steady horizontal wind pushes the sphere, holding it in equilibrium such that the string makes an angle of $30^\circ$ with the horizontal ceiling. The magnitude of the horizontal wind force is:

(1) $\frac{mg}{\sqrt{3}}$ (2) $\frac{1}{2}mg$ (3) $mg$ (4) $\sqrt{3}mg$

**Q40.**

A point light source is at the center of a spherical shell. If the shell's surface area increases 16 times, the maximum electric field amplitude on its surface will:

(1) decrease by 4 times
(2) decrease by 16 times
(3) increase by 4 times
(4) increase by 16 times

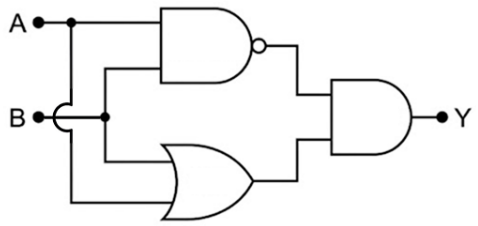
**Q41.**

In a Young's double slit setup, the slit separation is 1 mm and the screen is placed 1 m away. Using monochromatic light of wavelength 500 nm, the central maximum intensity is observed to be $I_m$. The minimum distance from the central maximum where the intensity drops to $I_m/2$ is:

(1) 0.250 mm
(2) 0.125 mm
(3) 0.500 mm
(4) 0.0625 mm

**Q42.**

Identify the correct truth table for a logic circuit:




(1)

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

(2)

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

(3)

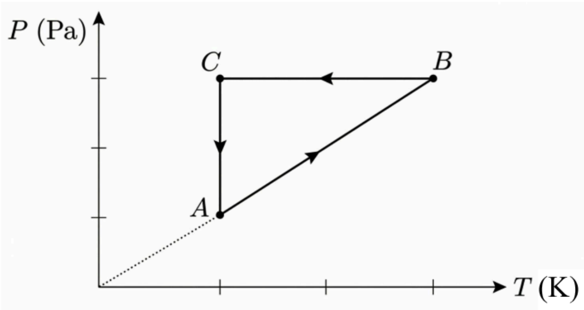
| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

(4)

| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

**Q43.**

An ideal gas undergoes a cyclic process ABCA shown in the P-T diagram. Path AB passes through the origin, BC is horizontal, and CA is vertical. Which of the following pairs of processes correspond to zero work done and zero change in internal energy, respectively?




(1) AB and BC (2) BC and CA
(3) AB and CA (4) CA and AB

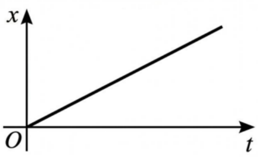
**Q44.**

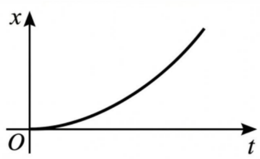
A converging lens forms a virtual image twice the object's size. The distance between the object and its image is 5 cm. The power of the lens is ____ D.

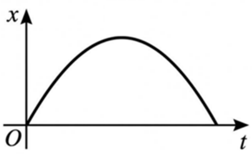
(1) $+5$ (2) $-10$ (3) $+10$ (4) $+20$

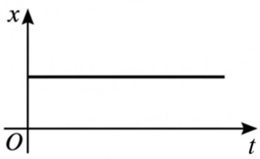
**Q45.**

The square of velocity ($v^2$) versus displacement ($x$) graph of a system is a straight line with a negative slope. If the system starts from the origin with a positive initial velocity, which of the following graphs describes its displacement ($x$) versus time ($t$) variation?

(1) 


(2) 


(3) 


(4) 

**Q46.**

A thin ring of radius 3 m carries a uniformly distributed charge $Q = 40\,\mu$C. A point charge $q = 5\,\mu$C is moved from the ring's center to a point on its axis 4 m away. The magnitude of work done by the electric field is ____ $\times 10^{-3}$ J. (Use $\dfrac{1}{4\pi\epsilon_0} = 9 \times 10^9$ N $\cdot$ m$^2$/C$^2$)

**Q47.**

An ideal monoatomic gas is compressed adiabatically so that its final volume becomes one-eighth of its initial volume. If the initial temperature of the gas is 200 K, the final temperature of the gas is ____ K.

**Q48.**

A uniform rod of mass 2 kg and length 3 m rotates at 10 rad/s about a central perpendicular axis. Two stationary beads of mass 0.5 kg each are gently placed at the two ends of the rotating rod and they stick to the rod. The final rotational kinetic energy of the system is ____ J.

**Q49.**

In a meter bridge experiment, a resistance of 10 Ω is connected in the left gap and an unknown resistance $S$ is connected in the right gap. The null point is obtained at a distance of 40 cm from the left end. When an additional resistance $X$ is connected in series with the resistance $S$ in the right gap, the null point shifts by 10 cm. The value of $3X$ (in Ω) is ____.

**Q50.**

A liquid drop of surface tension 0.25 N/m and radius 7 mm is broken into 64 identical smaller drops. If the work done in this process is $(400 + x)\ \mu$J, then the value of $x$ is ____. $(\pi = 22/7)$

**Q51.**

Consider the gaseous equilibrium $X_2(g) + Y_2(g) \rightleftharpoons 2XY(g)$ at temperature T. Initially, a 1 L closed vessel contains an equilibrium mixture of 1 mole each of $X_2$ and $Y_2$, and 2 moles of $XY$. If 1 mole of $XY$ is removed and the mixture is expanded to a 2 L volume at the same temperature, the number of moles of $X_2$, $Y_2$, and $XY$ at the new equilibrium, respectively, are:

(1) $0.75, 0.75, 1.50$
(2) $0.38, 0.38, 0.75$
(3) $1.25, 1.25, 0.50$
(4) $0.67, 0.67, 1.33$

**Q52.**

The spin-only magnetic moments for the following coordination complexes are in the increasing order:
(I) $[FeF_6]^{3-}$
(II) $[Co(C_2O_4)_3]^{3-}$
(III) $[NiCl_4]^{2-}$
(IV) $[Cr(NH_3)_6]^{3+}$

(1) II < III < IV < I
(2) II < IV < III < I
(3) I < IV < III < II
(4) III < II < IV < I

**Q53.**

Choose the INCORRECT statement regarding the general properties of group 13 and 14 elements:

(1) $BF_3$ acts as a stronger Lewis acid than $BCl_3$.
(2) The stability of the $+2$ oxidation state increases down group 14.
(3) Boron cannot form the $[BF_6]^{3-}$ complex ion.
(4) The atomic radius of gallium is less than that of aluminum.

**Q54.**

A double-stranded DNA segment contains a total of 20 bases. If exactly 4 of these bases are cytosine, what is the total number of hydrogen bonds in this segment?

(1) 20
(2) 28
(3) 24
(4) 30

**Q55.**

At a constant temperature, the vapor pressure of a pure liquid solvent is 150 torr. A solution prepared by dissolving 0.5 moles of a non-volatile weak electrolyte, $MX_2$ (yielding $M^{2+}$ and $X^-$ ions), into 4.0 moles of this solvent exhibits a vapor pressure of 120 torr. What is the degree of dissociation of $MX_2$?

(1) 0.30
(2) 1.00
(3) 0.50
(4) 0.20

**Q56.**

For which of the following species, first electron loss leads to a decrease in bond length and a diamagnetic product?

(1) $O_2$
(2) $NO$
(3) $N_2$
(4) $O_2^-$

**Q57.**

The unsaturated ether on acidic hydrolysis produces carbonyl compounds as shown below:

$CH_3 - CH = CH - O - CH = CH_2 \xrightarrow{H_3O^{(+)}} CH_3 - CH_2 - CH = O$ and $O = CH - CH_3$

Based on this cleavage pattern, predict the correct reagent that will successfully help to distinguish compounds "X" and "Y" obtained in the following reaction:

$(CH_3)_2C = CH - O - C(CH_3) = CH_2 \xrightarrow{H_3O^{(+)},\Delta} X + Y$

(1) Lucas reagent
(2) 2,4-DNP reagent
(3) Alkaline iodine solution
(4) Neutral $FeCl_3$ solution

**Q58.**

A chemistry student attempts the complete bromination of aniline by treating 9.3 g of the aniline with excess bromine water. After filtration and drying, the student recovers 24.75 g of the resulting white precipitate. What is the percentage yield of this reaction?

(1) 65.0%
(2) 75.0%
(3) 85.0%
(4) 92.5%

**Q59.**

Which cation pair is separated by selective precipitation using potassium chromate in dilute acetic acid?

(1) $Ba^{2+}$ and $Sr^{2+}$
(2) $Ba^{2+}$ and $Pb^{2+}$
(3) $Sr^{2+}$ and $Ca^{2+}$
(4) $Mg^{2+}$ and $Ca^{2+}$

**Q60.**

Find out the statements which are incorrect.

A. The acidic strength of terminal alkynes is greater than that of alkenes due to the higher s-character of the sp-hybridized carbon atom.

B. The $-NO_2$ group exerts a strong $+M$ (mesomeric) effect, which significantly activates the benzene ring towards electrophilic aromatic substitution.

C. A primary carbocation directly attached to an oxygen atom is highly stabilized by the resonance effect (back-bonding) of the oxygen's lone pairs.

D. Hyperconjugation in toluene stabilizes the molecule but directs incoming electrophiles predominantly to the meta position.

E. The inductive effect involves a permanent displacement of $\sigma$-electrons, whereas the electromeric effect is a temporary transfer of $\pi$-electrons.

Choose the correct answer from the options given below.

(1) A, B & C only
(2) B & D only
(3) C, D & E only
(4) B, D & E only

**Q61.**

0.5 moles of $\mathrm{I_2}$ (s) is added to 2 L of 1 M $\mathrm{Na_2S_2O_3}$ solution. After the reaction, the concentrations of $\mathrm{I^-}$, $\mathrm{S_4O_6^{2-}}$, and unreacted $\mathrm{S_2O_3^{2-}}$ are respectively (assume volume remains constant):

(1) 0.5 M, 0.25 M, 0.5 M
(2) 1.0 M, 0.5 M, 1.0 M
(3) 0.25 M, 0.5 M, 0.5 M
(4) 0.5 M, 0.5 M, 0 M

**Q62.**

The correct increasing order of the magnitude of negative electron gain enthalpy for N, O, S, and Cl is:

(1) $N < O < S < Cl$
(2) $O < N < S < Cl$
(3) $N < S < O < Cl$
(4) $N < O < Cl < S$

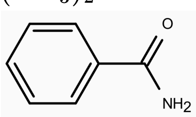
**Q63.**

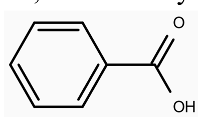
The wavenumber of the spectral line observed in the spectrum of $He^+$ ion, when an electron transitions between two levels whose product is 6 and sum is 5, is:

(1) $\frac{5R}{36}$
(2) $\frac{5R}{9}$
(3) $\frac{7R}{9}$
(4) $\frac{4R}{9}$

**Q64.**

Given below are two statements:
Statement I: In the gas phase, the basic strength of aliphatic amines follows the order $3^\circ > 2^\circ > 1^\circ > NH_3$. However, in aqueous solution, the basicity order for methyl-substituted amines is $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2 > NH_3$.

Statement II: 

undergoes Hoffmann bromamide degradation with $Br_2/NaOH$ to produce compound (X). Compound (X) reacts with $NaNO_2/HCl$ at 273-278 K, followed by treatment with $CuCN/KCN$ to give compound (Y). Complete hydrolysis of compound (Y) yields 

.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are false
(2) Both Statement I and Statement II are true
(3) Statement I is true but Statement II is false
(4) Statement I is false but Statement II is true

**Q65.**

From the following IUPAC names, how many compounds contain at least one tertiary alcohol?
(I) 2-Methylbutan-2-ol
(II) 3-Methylpentan-2-ol
(III) 2,3-Dimethylbutan-1,2-diol
(IV) Cyclohex-2-en-1-ol
(V) 1-Methylcyclopentanol
(VI) 2,2-Dimethylpropan-1-ol
Choose the correct answer from the options given below:

(1) Four (2) Three (3) Two (4) Five

**Q66.**

Given below are two statements:
Statement I: Both acetaldehyde and benzaldehyde reduce Fehling's solution to yield a red precipitate of cuprous oxide.
Statement II: When a mixture of benzaldehyde and formaldehyde is heated with concentrated NaOH, formaldehyde is oxidized to the formate ion due to its higher reactivity towards nucleophiles.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are false
(2) Statement I is false but Statement II is true
(3) Both Statement I and Statement II are true
(4) Statement I is true but Statement II is false

**Q67.**

A gas mixture of $40\%$ $He$ and $60\%$ $CH_4$ by mole is equilibrated with water at a total pressure of 10 atm. If the Henry's law constants $(K_H)$ for $He$ and $CH_4$ are $1.6 \times 10^5$ atm and $4.0 \times 10^4$ atm respectively, what is the ratio of their dissolved mole fractions $(x_{He} : x_{CH_4})$ in the aqueous solution?

(1) 1 : 6 (2) 1 : 4 (3) 2 : 3 (4) 8 : 3

**Q68.**

The ionization enthalpy of $He^+$ and the work function of a metal are '$x$' and '$y$' kJ mol$^{-1}$ respectively. The maximum kinetic energy of a photoelectron ejected by the light emitted during the $n = 2$ to $n = 1$ transition in $He^+$ is expressed in SI units as:

(1) $\frac{1000(3x-y)}{4N_A}$
(2) $\frac{N_A}{250(3x-4y)}$
(3) $\frac{250(3x-4y)}{N_A}$
(4) $\frac{1000(x-y)}{N_A}$

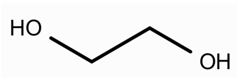
**Q69.**

"Y" is a yellow oxoanion of the lightest element of group 6. In an acidic medium, "Y" converts into another oxoanion "Z". The color of "Z" and the number of M-O-M bridging bonds in "Z" respectively are:

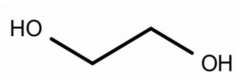
(1) orange, one
(2) orange, two
(3) yellow, one
(4) green, zero

**Q70.**

Given below are two statements:

Statement I: For 

, the gauche conformer is more stable than the anti conformer despite having a smaller dihedral angle between the hydroxyl groups.

Statement II: The anti conformer of 

possesses higher torsional strain than the gauche conformer, making it less stable.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are false
(2) Statement I is false but Statement II is true
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are true

**Q71.**

0.5 g of an organic compound was analysed using Kjeldahl's method for the estimation of nitrogen. The ammonia gas evolved was completely absorbed in 50 mL of 0.5 M $H_2SO_4$. The residual acid required 30 mL of 1 M NaOH for complete neutralization. The percentage of nitrogen in the given organic compound is ____.
(Given: atomic mass of N = 14)

**Q72.**

Two radioactive isotopes, P and Q, have half-lives of 10 minutes and 15 minutes, respectively. Initially, they contain the same number of atoms. After $x$ minutes, the ratio of atoms of Q to P becomes 8. The value of $x$ is ____.

**Q73.**

The specific conductivity of a saturated solution of a sparingly soluble salt $AB_2$ at 298 K is $2.5 \times 10^{-5}$ S cm$^{-1}$. The specific conductivity of the water used to prepare this solution is $0.5 \times 10^{-5}$ S cm$^{-1}$. If the limiting molar conductivities of $A^{2+}$ and $B^{-}$ ions are 120 and 40 S cm$^2$ mol$^{-1}$ respectively, the solubility product ($K_{sp}$) of the salt is $x \times 10^{-12}$. The value of $x$ is ____.

**Q74.**

A coordination complex of Cobalt(III) has the empirical formula $\mathrm{CoCl_3 \cdot 5NH_3 \cdot H_2O}$. The molar conductivity of its aqueous solution corresponds to that of a $1:2$ electrolyte. When an aqueous solution containing $0.02$ moles of this complex is treated with an excess of aqueous $\mathrm{AgNO_3}$, $X$ millimoles of a white precipitate is formed. Given that the spin-only magnetic moment of the central metal ion in this complex is $Y$ Bohr Magnetons (BM), determine the numerical value of $(X + Y)$.

**Q75.**

A gaseous alkene (P) decolorizes bromine water. 21 g of (P) occupies 11.2 dm$^3$ at STP (molar volume $= 22.4$ dm$^3$). Acid-catalyzed hydration of (P) yields a major product (Q), which upon oxidation forms a ketone (R). (R) reacts with $CH_3MgBr$ followed by hydrolysis to give compound (Z). The molar mass of (Z) is ____ g/mol.

## ANSWERS KEY

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **1.** (1) | **2.** (2) | **3.** (3) | **4.** (2) | **5.** (1) | **6.** (4) | **7.** (3) | **8.** (2) |
| **9.** (2) | **10.** (3) | **11.** (2) | **12.** (1) | **13.** (3) | **14.** (2) | **15.** (3) | **16.** (2) |
| **17.** (2) | **18.** (2) | **19.** (2) | **20.** (3) | **21.** 20.0 | **22.** 4.0 | **23.** 113.0 | **24.** 289.0 |
| **25.** 1014.0 | **26.** (2) | **27.** (2) | **28.** (2) | **29.** (3) | **30.** (3) | **31.** (2) | **32.** (2) |
| **33.** (3) | **34.** (2) | **35.** (3) | **36.** (2) | **37.** (2) | **38.** (2) | **39.** (4) | **40.** (1) |
| **41.** (2) | **42.** (3) | **43.** (3) | **44.** (3) | **45.** (3) | **46.** 240.0 | **47.** 800.0 | **48.** 30.0 |
| **49.** 25.0 | **50.** 62.0 | **51.** (1) | **52.** (1) | **53.** (1) | **54.** (3) | **55.** (3) | **56.** (2) |
| **57.** (3) | **58.** (2) | **59.** (1) | **60.** (2) | **61.** (1) | **62.** (1) | **63.** (2) | **64.** (4) |
| **65.** (2) | **66.** (2) | **67.** (1) | **68.** (3) | **69.** (1) | **70.** (3) | **71.** 56.0 | **72.** 90.0 |
| **73.** 4.0 | **74.** 40.0 | **75.** 74.0 | | | | | |